

LE CHUNG HIEU

DIGITAL IC DESIGN INTERN

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PERSONAL STATEMENT

Motivated final-year Electronics Engineering student specializing in Digital IC Design with hands-on experience in RTL development, FPGA prototyping, and hardware accelerator implementation using Verilog/SystemVerilog. Experienced in developing processor architecture, cryptographic hardware, and DSP systems, including RV32I RISC-V processor, AES-CTR accelerator, CRYSTALS-Dilithium PQC implementation, and real-time FPGA-based signal processing systems. Familiar with RTL design flow including functional verification, simulation, waveform debugging, synthesis, and FPGA deployment using industry-standard EDA tools. Seeking an internship opportunity in Digital IC / ASIC Design to further develop expertise in RTL Design, ASIC frontend flow, and SoC development.

EDUCATION

Ho Chi Minh City University of Technology (VNU-HCMUT)	2022 – Current
Final-year student	GPA: 7.1/10
Faculty: Electrical and Electronics Engineering	Major: Electronics and Telecommunications Engineering

LANGUAGES

TOEIC (LISTENING/READING): 790/990

PROJECTS

COMPUTER ARCHITECTURE (INSTRUCTOR: ASSC. PROF. DR. TRAN HOANG LINH)

Ho Chi Minh City University of Technology	Oct 2025 – Dec 2025
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Technologies: SystemVerilog, Cadence Xcelium, Quartus, DE2-115 Altera FPGA Kit.

Description: Designed and implemented a 32-bit RV32I RISC-V processor in SystemVerilog following a modular RTL architecture with separated datapath and control logic to improve scalability and maintainability. The processor was developed using a 5-stage pipelined architecture (IF, ID, EX, MEM, WB) to increase instruction throughput and support efficient execution flow. Implemented major RTL components including ALU, Register File, Control Unit, Immediate Generator, Program Counter, and Memory Interface. Designed hazard detection logic, forwarding paths, and pipeline control mechanisms to resolve data dependencies and pipeline hazards during instruction execution. The design was synthesized and deployed on the DE2-115 FPGA platform, successfully executing application programs such as stopwatch implementation.

CRYSTALS-DILITHIUM PQC SIGNATURE - FPGA IMPLEMENTATION (INSTRUCTOR: ASSC. PROF. DR. TRAN HOANG LINH)

Ho Chi Minh City University of Technology	Oct 2025 – Dec 2025
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Technologies: Verilog, VHDL, Vivado, Quartus, ModelSim.

Description: Implemented hardware acceleration modules for the CRYSTALS-Dilithium post-quantum cryptographic signature scheme on FPGA with emphasis on NTT-based polynomial arithmetic acceleration and datapath optimization. The design focused on improving computational efficiency while reducing hardware resource utilization. Developed RTL modules including NTT/Inverse NTT engine, butterfly processing elements, modular reduction blocks, and address generation logic to support polynomial operations required by the Dilithium algorithm. Optimized the datapath structure and memory access organization to improve performance and FPGA implementation efficiency. Verified functionality through RTL simulation, waveform analysis, and test vector validation using ModelSim. The hardware implementation was synthesized on FPGA and evaluated through timing and resource utilization analysis.

FPGA-BASED AES-CTR ENCRYPTION CORE (INSTRUCTOR: M.SC. NGUYEN TUAN HUNG)

Ho Chi Minh City University of Technology

Jun 2025 – Aug 2025

Technologies: SystemVerilog, Vivado, Quartus.

Description: Designed and implemented a hardware encryption accelerator based on AES operating in Counter (CTR) mode, supporting AES-128, AES-192, and AES-256 key configurations. The architecture was developed using a modular RTL design approach to improve reusability and simplify verification. Implemented key hardware blocks including KeyExpansion, Cipher Engine, CTR control logic, and counter generation mechanism. The datapath and control architecture were optimized to improve throughput while maintaining hardware efficiency for FPGA deployment. Performed functional verification through testbench simulation and waveform analysis using ModelSim and Vivado Simulator. The implementation was validated using encryption test vectors and successfully deployed on the DE1-SoC FPGA platform to verify real hardware operation.

DIGITAL SIGNAL PROCESSING ON FPGA (INSTRUCTOR: M.SC. NGUYEN TUAN HUNG)

Ho Chi Minh City University of Technology

Jan 2025 – May 2025

Technologies: SystemVerilog, Quartus, ModelSim.

Description: Designed and implemented FIR and IIR digital filters using RTL design techniques for real-time audio processing applications on FPGA. The project focused on improving filter stability, hardware efficiency, and real-time processing capability. Developed filter architectures and optimized arithmetic structures to reduce hardware cost while maintaining signal quality. Integrated an I2C-based audio codec interface to support real-time audio acquisition and playback functionality. Performed simulation, functional verification, waveform analysis, and timing evaluation using ModelSim and Quartus to validate system operation and hardware behavior.

SKILLS

- **Hardware Description Languages:** SystemVerilog, Verilog, VHDL.
- **Digital IC Design:** RTL Design, FSM Design, Datapath and Control Architecture, Pipeline Design, RISC-V Architecture, DSP Hardware Design, Cryptographic Accelerator Design.
- **Verification:** Testbench Development, Self-checking Testbench, Functional Verification, Waveform Debugging.
- **EDA Tools:** Quartus, Vivado, ModelSim, Cadence Xcelium, Visual Studio Code.
- **FPGA Platforms:** DE1-SoC, DE2-115, DE10.
- **Soft skills:** Teamwork, Communication Skills, Problem analysis.